

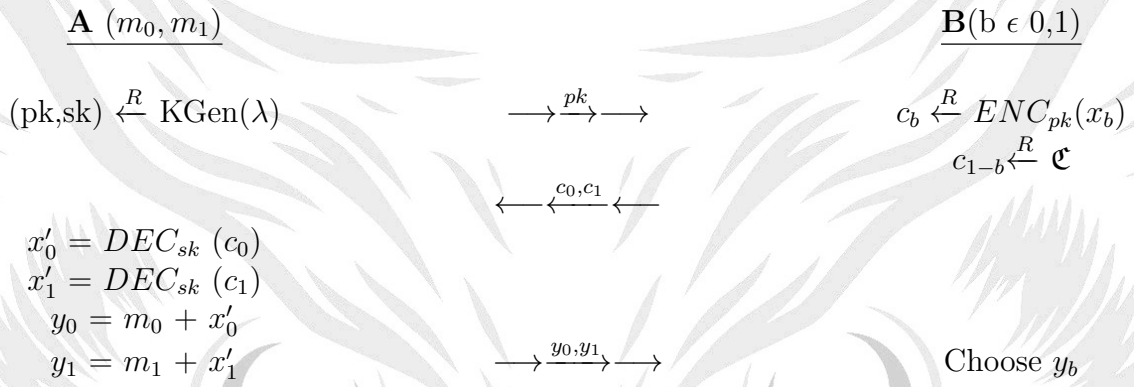
Oblivious Transfer

Pushpendra Pal

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1 Micali OT

1.1 Protocol



1.2 Questions

- What if B intentionally sends a valid c_{1-b} , then can he not get m_1 too?

2 Bellare-Micali OT

2.1 Protocol

A (m_0, m_1)

$c \xleftarrow{R} \mathcal{C}$

$\longrightarrow \xrightarrow{c} \longrightarrow$

$\longleftarrow \xleftarrow{y_0, y_1} \longleftarrow$

- IF $y_0 \cdot y_1 = c$; do {

- $r_0, r_1 \xleftarrow{R} Z_p$
- $c_0 = (g^{r_0}, [\mathbf{H}(y_0^{r_0}) \cdot m_0])$
- $c_1 = (g^{r_1}, [\mathbf{H}(y_1^{r_1}) \cdot m_1])$ }

- ELSE ABORT

$\longrightarrow \xrightarrow{c_0, c_1} \longrightarrow$

B($b \in \{0,1\}$)

$k \xleftarrow{R} Z_p, \text{ord}(g) = p$

$y_b = g^k$

$y_{1-b} = c \cdot g^{-k}$

- Take c_b , let's say $b=0$ then $(v_0, v_1) = c_0$.

– $v_0 = g^{r_0}$

– $v_1 = \mathbf{H}(y_0^{r_0}) \cdot m_0$

- Since $(v_0)^k = (g^{r_0})^k = (g^k)^{r_0} = (y_0)^{r_0}$

- Find $\mathbf{H}(v_0^k) = \mathbf{H}(y_0^{r_0})$,

- Find $\mathbf{H}(y_0^{r_0}) \cdot v_1 = \mathbf{H}(y_0^{r_0}) \cdot [\mathbf{H}(y_0^{r_0}) \cdot m_0] = m_0$

2.2 Queries

- Explain why can't B get m_1 if $b = 0$, given that he knows k and c ?

Ans. Let's do $(g^{r_1})^{-k}$, then find c^{r_1} , but B doesn't know r_1 , and so the problem is in a way a DLog problem.